

Data Management

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(Please start your computers)





[Disclaimer]

- This training offers general information about data management.
- Although most of these principles apply to all institutes, I work at JIC, so the content of the training may not align 100% with your institute's policies.
- I encourage you to follow up with your bioinformatics teams and/or with your line manager at your institute to learn more about data management and the institute's specific policies.
- This is the JIC document for Procedures and Best Practices:

<https://intranet.nbi.ac.uk/apps/documents/download.php?docID=66239>



[Presentation available at:]

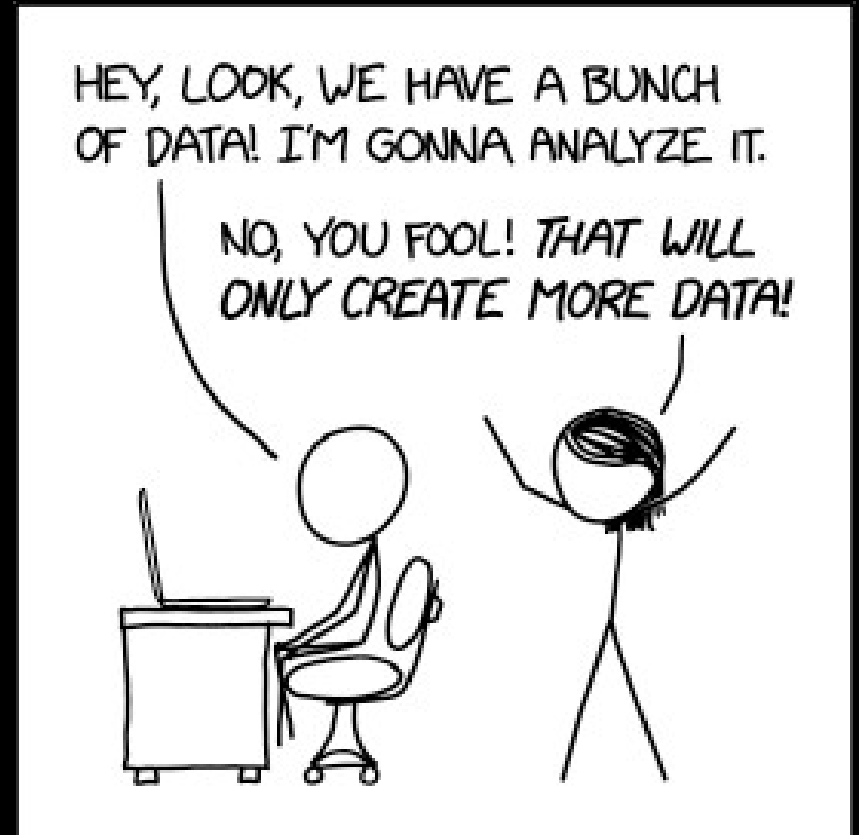
–<https://repo.informatics.jic.ac.uk/>

JIC Informatics Repo			 John Innes Centre	
Name	Last modified	Size		
 ai-python26/	2026-04-30 09:25	-		
 cmd_line_shell/	2026-04-13 16:13	-		
 Data_Management_2026/	2026-05-20 12:36	-		
 GIS_course_2025/	2025-05-13 11:50	-		
 GIS_course_2026/	2026-05-19 17:09	-		
 Miscellany/	2026-05-20 12:36	-		



[Course Content]

- Part 1: Theory and open discussion.
 - Data ownership and governance
 - FAiR and AI-ready data
 - Filesystems
 - Bad practices
 - Metadata
 - Instrument network
 - Recovering data
 - Sharing data
- Part 2: Hands-on practice.
 - Command Line file manipulation
 - File integrity
 - Big data
 - Uploading data to remote repositories



[Reflections]

- Who are the stakeholders of data?
- Who owns data?



[Who are the stakeholders of data]

- Researcher (PI/GL, postdoc, PhD student).
- Institution (JIC/EI/QIB/TSL).
- Collaborators.
- Funders (UKRI/BBSRC/Industry).
- Publishers.
- The taxpayer.
- Humanity.



[Data Ownership]

- The Institute.
- The Funder (BBSRC requires data retention for a minimum of 10 years).
- Intellectual Property.
- Commercial Partners.
- Collaborators.

The responsibility for maintaining and securing data always resides with the Project Leader/GL.



[As a user]

- Ensure the data is available, secure, correct, understandable, reproducible, defensible.
- The other stakeholders of the data you create, and work with, will have their own policies on how data should be treated.
- UKRI Outline Data Management Plan
 - <https://www.ukri.org/councils/nerc/guidance-for-applicants/what-to-include-in-your-application/outline-data-management-plan/>



[GDPR and Human data]

- General Data Protection Regulation (GDPR)
 - Regulates the collection and processing of personal data from individuals.
 - <https://gdpr-info.eu>
- Genetic data is a "special category" under GDPR.
 - Guidelines from your institute/funder.
 - Speak to your data manager.
 - Read your data management plan.
 - Special considerations for publishing and anonymizing human data.



[FAIR (Findable, Accessible, Interoperable and Re-usable)]

1. Unified Data Architecture

- Standardised column names across all experiments
- Vendor outputs converted to common industry units
- Controlled vocabularies for genes, cell lines, and samples

2. Contextual Metadata Frameworks

- Capture experimental context and provenance
- Include baseline settings and environmental conditions
- Enable direct integration with public data platforms
- Maintain full traceability from raw data to analysis

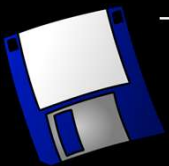
3. Shared Computational Workflows

- Replace isolated scripts with shared executable notebooks
- Build reproducible and versioned analysis pipelines
- Automate submission to public repositories
- Register workflows in centralised workflow registries



[AI-Ready Data]

- AI models cannot guess context. Data must be structured so algorithms can automatically parse, clean, and link it at scale.
- Key Criteria for AI-Ready Data
 - Open, machine-readable formats → (JSON, CSV, Parquet instead of PDFs or proprietary files).
 - Harmonised features and units → Consistent column names and standardised measurements across datasets.
 - Rich metadata and provenance → Explicitly capture experimental variables, instrument settings, and environmental conditions.
 - High data integrity → Remove duplicates, undocumented gaps, and unclear file naming conventions.
 - Scalable interoperability → Data designed for automated integration with workflows, repositories, and AI pipelines.

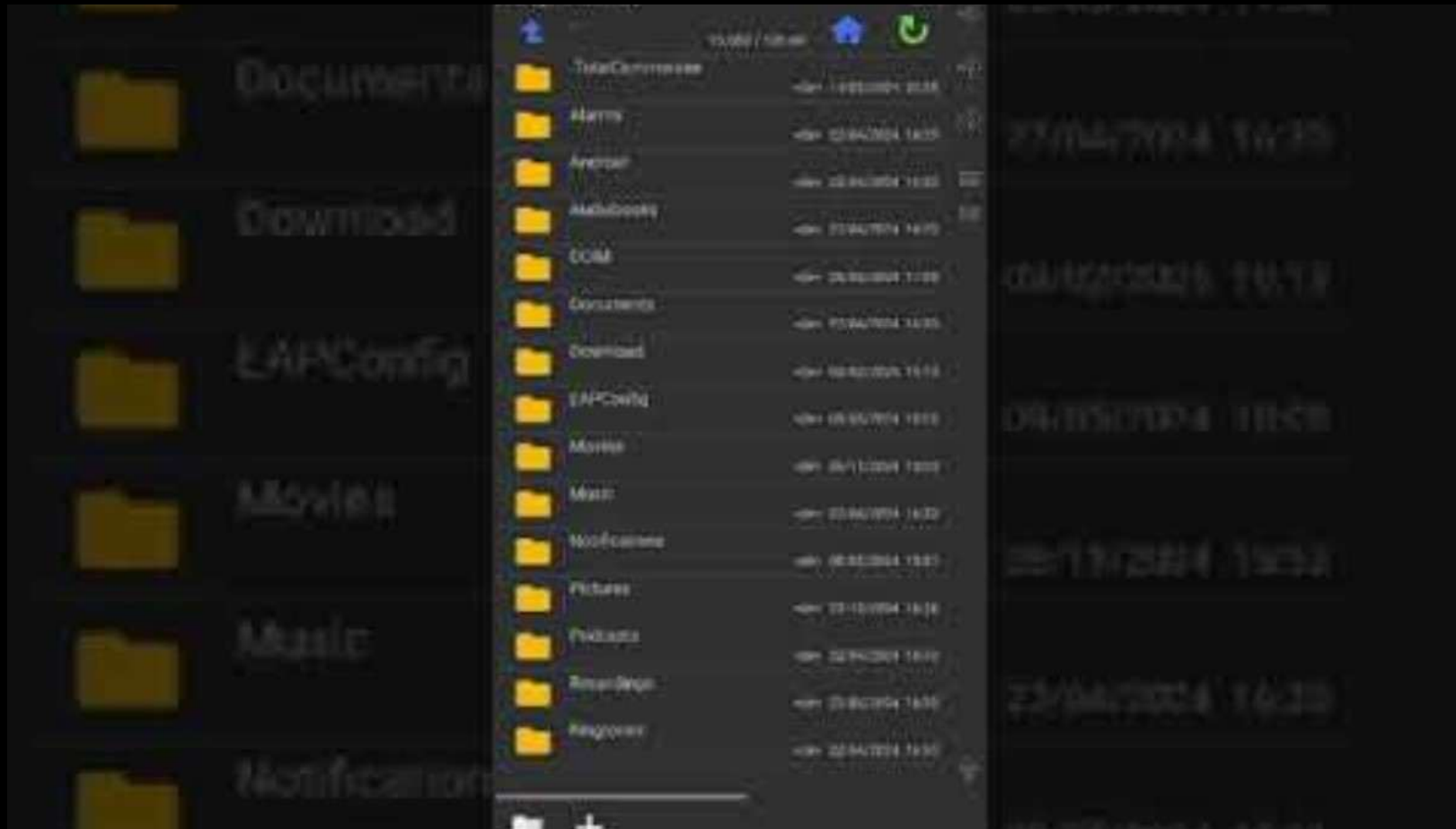


[What is a filesystem?]

- Is what we use to organise and store data on hardware devices like hard drives, SSDs, and network storage.
- Manages how data is written, read, and accessed efficiently.
- Uses a hierarchical structure of directories (folders) and files.
- Provides access control and permissions, defining who can read, write, or execute files.
- Different types exist, optimised for different use cases.
- In Windows, you use \ to separate the folders; in Linux and macOS, you use /



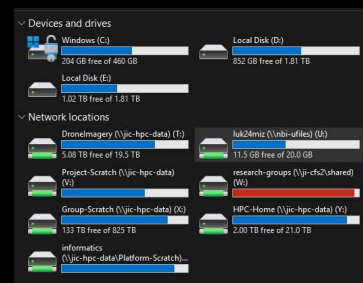
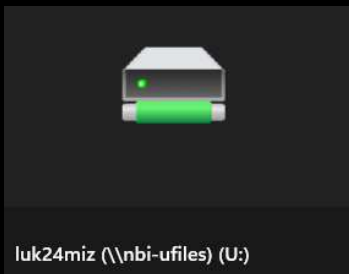
[Example: The file system behind android]



[Storage locations available at the NBI]

HERE YOUR DATA
IS NOT SAFE

- U drive.
- OneDrive and SharePoint.
- Computing network shared storage [\\nbi-cfs2](#)
- HPC storage (Also known as Isilon). [\\jic-hpc-data](#)



[U drive and Computing shared locations]

- The U drive
 - Can store up to 20G
 - Good for documents that do not require sharing
 - Fully backed up and snapshots
 - Automatically mapped
 - Access: [\\nbi-ufiles\username](#)
- Computing shared folders
 - Quota varies but is small, not designed for big data.
 - Good for shared documents like protocols, SOPs
 - Fully backed up and snapshots
 - Needs manual mapping
 - Access: [\\files](#) [\\ji-cfs2](#)



[OneDrive – SharePoint]

- Good for sharing drafts, writing protocols, and collaborating on office documents with colleagues.
- Fully backed up, secure, and officially supported by NBI Computing.
- Ideal for real-time co-authoring and managing team file access across internal JIC/NBI groups.
- IMPORTANT:
 - Do not use it to store or sync raw bioinformatics data or large file trees; it will break metadata, choke bandwidth, and trigger endless sync conflicts.
 - Be organised, create folders and compartmentalise your data; it is easy to have all the data in the root of your OneDrive folder.



[Omero – Image Storage]

- Managing, viewing, and organising microscopy and high-dimensional biological images.
- Centralised NBI storage that preserves data integrity, avoids local loss, and is fully backed up.
- Easily share datasets, annotations, and metadata across internal groups without copying huge files.
- Do not use standard filesystems or personal cloud drives for images; they strip away essential acquisition metadata and break traceability.

omero.nbi.ac.uk

(ask your bioinformatics team or microscopy)



[Data Access/Permissions]

- At the JIC, Group Leaders have private spaces at Computing's Shared folders and the HPC.
- Group members can then get access.
- Speak to informatics for more complex arrangements.
 - Collaborators' access.
 - Project storage across groups/members.
 - Institute-wide resource.
 - External access.
- It may not be the case for your institute; consult your line manager or your bioinformatics teams to learn more.



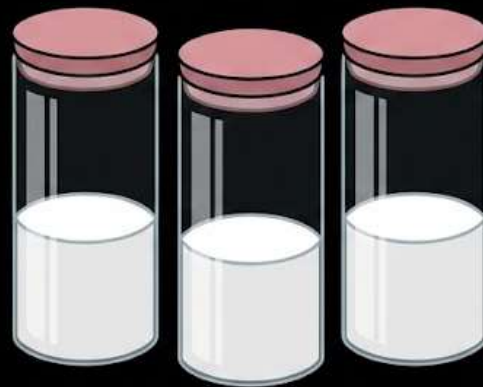
[Bad practices]

- Local-Only Storage. Storing data solely on external drives or laptops risks total loss (e.g., hardware failure or theft). Use our institutional and backed-up storage systems.
- Editing Raw Data. Working directly on raw files endangers data integrity and reproducibility. Make copies; treat raw data as read-only.
- No Backups. A single copy is a single point of failure. Automate regular backups in multiple locations. A snapshot is not a backup!
- Running analysis in Group Folders. Use JIC-Scratch for heavy I/O!
- Disorganized Files, Cryptic filenames like new_final2_real.bam confuse everyone—including you. Use clear, consistent naming conventions and folder structures.
- No Metadata or Documentation, Unlabelled data becomes unusable. Document sample origins, processing steps, and tool versions.



[Metadata]

UNLABELED DATA IS JUST
A MYSTERY JAR



NO METADATA



DATA WITH METADATA



[Metadata]

- Your final results are only as useful as your documentation!
- Documenting your analysis ensures reproducibility, collaboration, and future understanding.
- Best Practices:
 - Store your analysis scripts (e.g., Python, R, Bash)
 - Include notes on:
 - Data sources and versions.
 - Parameters and tools used.
 - Rationale for choices and any caveats.
- Use Tools Like Git, GitLab or GitHub for version control and collaboration.
- Be sure that your metadata is good before you set your data to be archived

You will be the first person to find your descriptions of the data useful.



[Metadata – template]

- https://repo.informatics.jic.ac.uk/Data_Management_2025/Metadata_template.txt
 - General Information.
 - Sample Information.
 - Experimental Information.
 - File Information.
 - Ethical & Access Information.
- Customise it or create your own template; check the requirements of the platforms where your data will be uploaded for guidance, and remember, we are here to help.

ASK your bioinformatics team, different institutes have different policies.



[Metadata – template]

```
description: [REDACTED] mapping data - part of the Antirrhinum hybrid zone project
project: Evolutionary Dynamics Underlying Species Diversification
confidential: false
personally_identifiable_information: false
owners:
  - name: [REDACTED]
    email: [REDACTED]
    username: [REDACTED]
creation_date: 2018-03-29
species: Antirrhinum pseudomajus/striatum
note: This is a temporary overflow archive. At some future point we will retrieve it and clean up derived files, etc. but time and space are tight!
# links:
# - http://doi.org/your\_doi
# - http://github.com/your\_code\_repository
# budget_codes:
# - E.g. [REDACTED]
```



[The Instrument Network]

- Some computers connected to instruments like microscopes are not fully connected to the NBI network because they do not comply with basic security requirements, such as regular updates, a supported operating system, and antivirus software.
- Moving data around on USB sticks is neither efficient nor safe; I discourage it.
- Because of this, all these instruments have access to NBI's central storage and can be used to upload raw data and results to your group's shared location or personal folder, following the steps from the previous slides.
- You need to contact computing.helpdesk@nbi.ac.uk:
 - If the instrument you work with is not connected to the Instrument Network.
 - If you need remote access to an instrument from your institute workstation. This can be arranged by contacting the helpdesk.

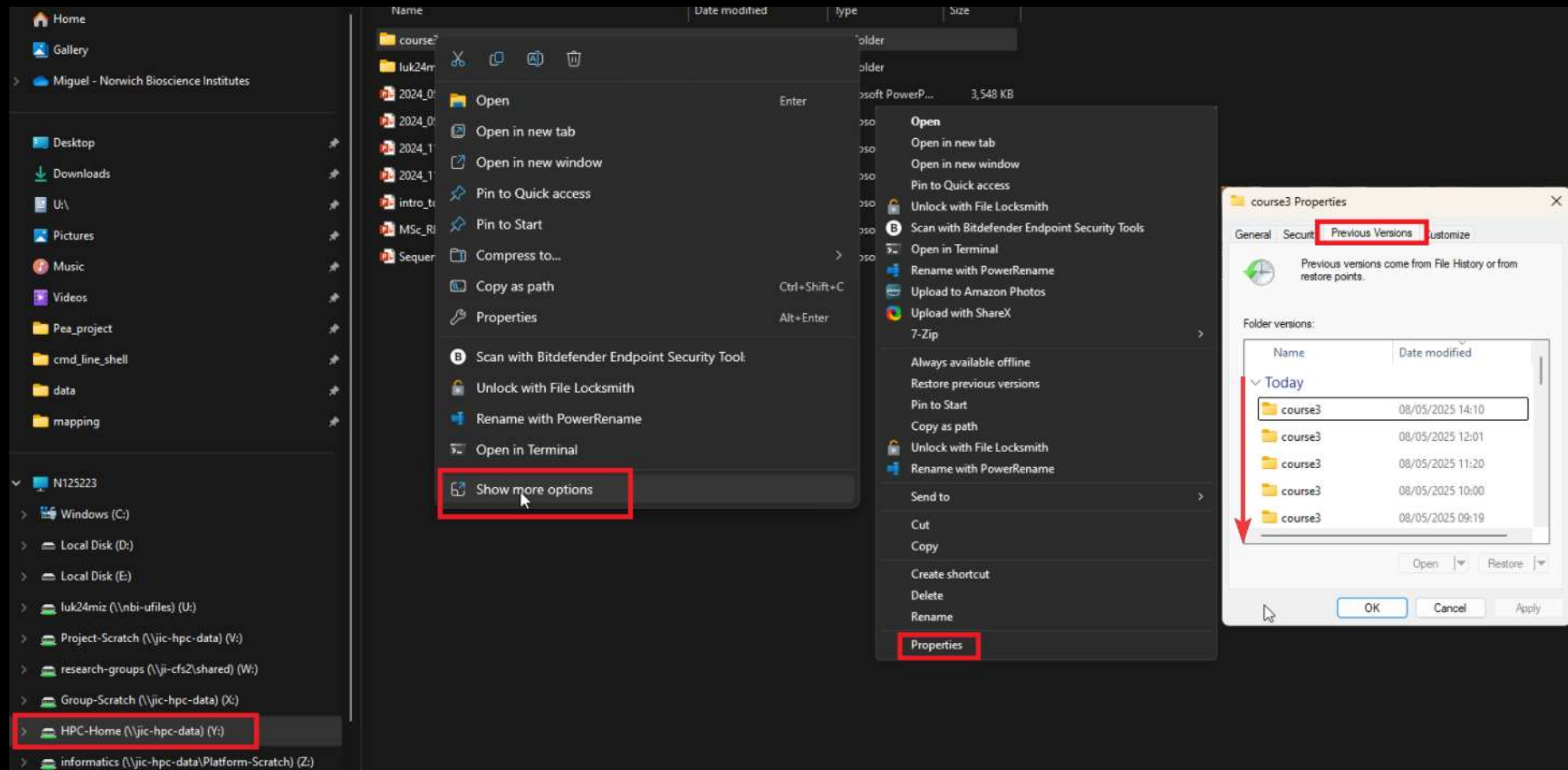


[How to recover data, if possible]

- The Isilon automatically takes snapshots of the data. However, depending on the folder, the number of snapshots and the retention policies may vary.
- Computing's locations and OneDrive also have retention policies.
- If you have recently deleted (or modified a file), the chances of recovering it are high. The more time passes, the lower the chances of getting it back.
- Permissions apply identically to the snapshots as they did to the original data.
- Snapshots can be accessed from any device.
- The best way of understanding what is available is to navigate through the snapshots



[Recovering data from Windows]



[Recovering Exercise]

- Explore the different snapshots using your Windows device.
- Compare the snapshot folder with the “current” folder.
- Recover some data from any of the snapshots (be careful and do not overwrite anything).



[NIB Tools for sharing data]

Alongside OneDrive, which is recommended only for sharing small amounts of data, NBI provides two dedicated data transfer services:

Dropoff: Easy-to-use web-based transfer service for datasets up to 300 GB.

<https://dropoff.nbi.ac.uk/>

Globus: High-performance data transfer platform with no practical data size limit. Ideal for large datasets and automated transfers.

<https://www.globus.org/>

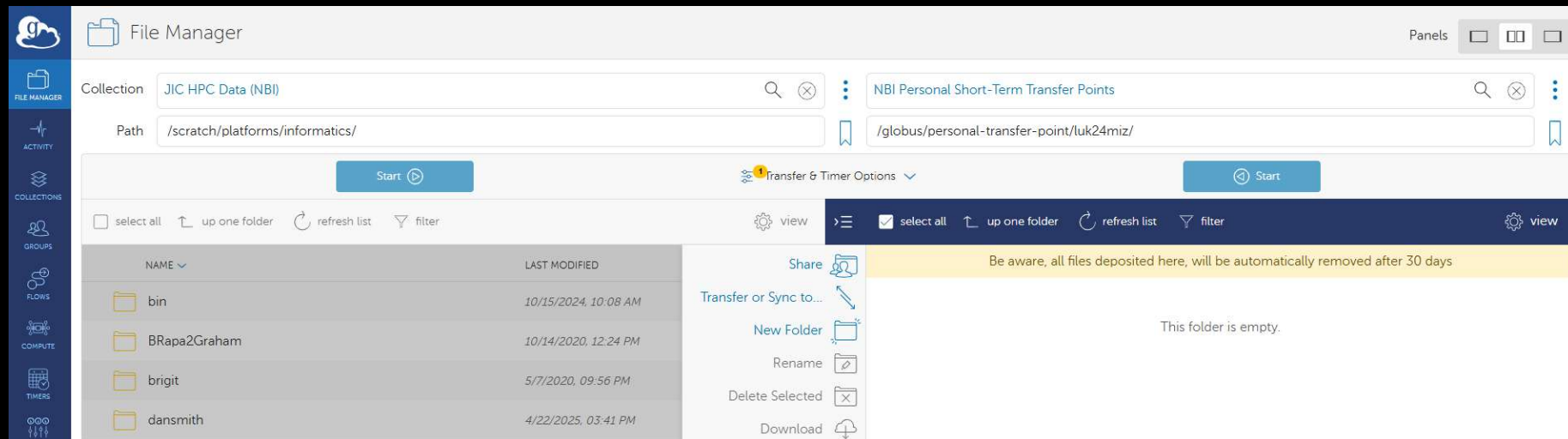
Choose the right tool:

- Small files → OneDrive
- Up to 300 GB → Dropoff
- Large datasets or recurring transfers → Globus



[Globus]

- Globus now has access to the JIC storage, collection: **JIC HPC Data (NBI)**.
- User's shared collection are in **NBI Personal Short-Term Transfer Points**.
- You can create your own guest collections and share with your collaborators.



[Globus – how to share]

- In Globus, data is organised and accessed through collections.
- You cannot share data directly from the HPC storage; it must first be moved to the external endpoint.
- You can create a guest collection to share data with your collaborators.
- Detailed instructions are available in the Research Computing documentation.

[Globus - Research Computing Support – Confluence](#)



[Git – protect your scripts]

- Keeps a history of changes
- Lets you go back to an older version
- Protects against accidental changes/deletion
- Makes it easier to share and collaborate
- Records who changed what and when

Edit script

↓

git add

↓

git commit

↓

Git repository

↓

Safe version history

Go to your scripts directory

cd my_project

Start Git

git init

Add your script

git add my_script.py

Save a version

git commit -m "Initial version"

Connect to your remote repository

git remote add origin

https://github.com/USERNAME/my_project.git

Push to the remote repository

git branch -M main

git push -u origin main



[Working Remotely]

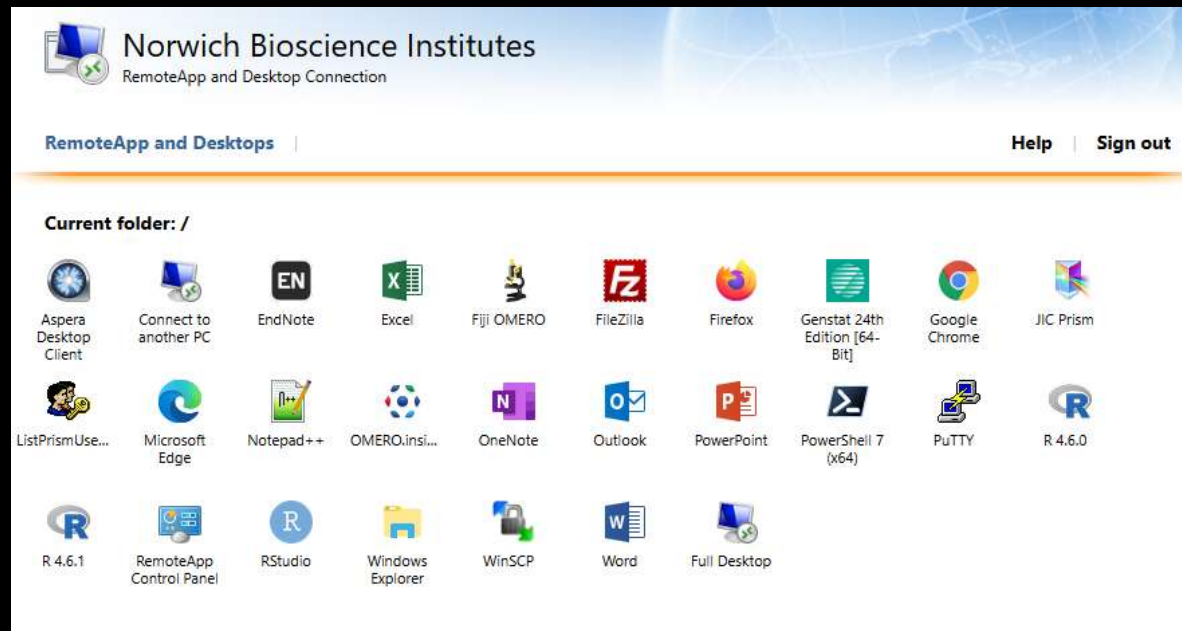
- Remote Access & VPN
 - VPN: Works from NBI and non-NBI devices.
 - NBI device: Nearly full network access; folders can be mapped.
 - Non-NBI device: Restricted access; no folder mapping and limited VM/server access.
 - Both: Remote Desktop and `winrds.nbi.ac.uk` available.
- WFH recommendation:
If possible, remote into an office computer – better connection to NBI resources and less risk of losing work due to disconnections.



[Working Remotely]

winrds.nbi.ac.uk

- Provides Remote Desktop through a temporary Windows VM.
- Access to most NBI resources.
- Volatile: the VM is temporary; don't store data or work locally.
- Ideal if you don't have an office PC to remote into.



[Sharing Exercise]

- Upload some data into Dropoff and share it with your personal email account. Download the data using your personal email to oversee the full process.
- Create a file with your username in your U drive, uploaded into a new folder in Globus and share it with me luk24miz@nbi.ac.uk
- Explore your access. Create a folder in your external NBI short-term endpoint and transfer some data from your institute's HPC endpoint



[Scientific Storage Locations for JIC Users]

- Archive
 - Only data champions and JIC Informatics members can add data to it.
 - Same access speed as Scratch.
 - What you should store here:
 - Annotated Raw Data.
 - Results.
- Home
 - Your personal space 300GB
 - What you should store here:
 - Environments.
 - Software (backed up by version control: Git, GitHub, GitLab).
- Scratch
 - Only temporary storage; this data is not backed up
 - What you should store here:
 - Expendable data.
 - Intermediate files.

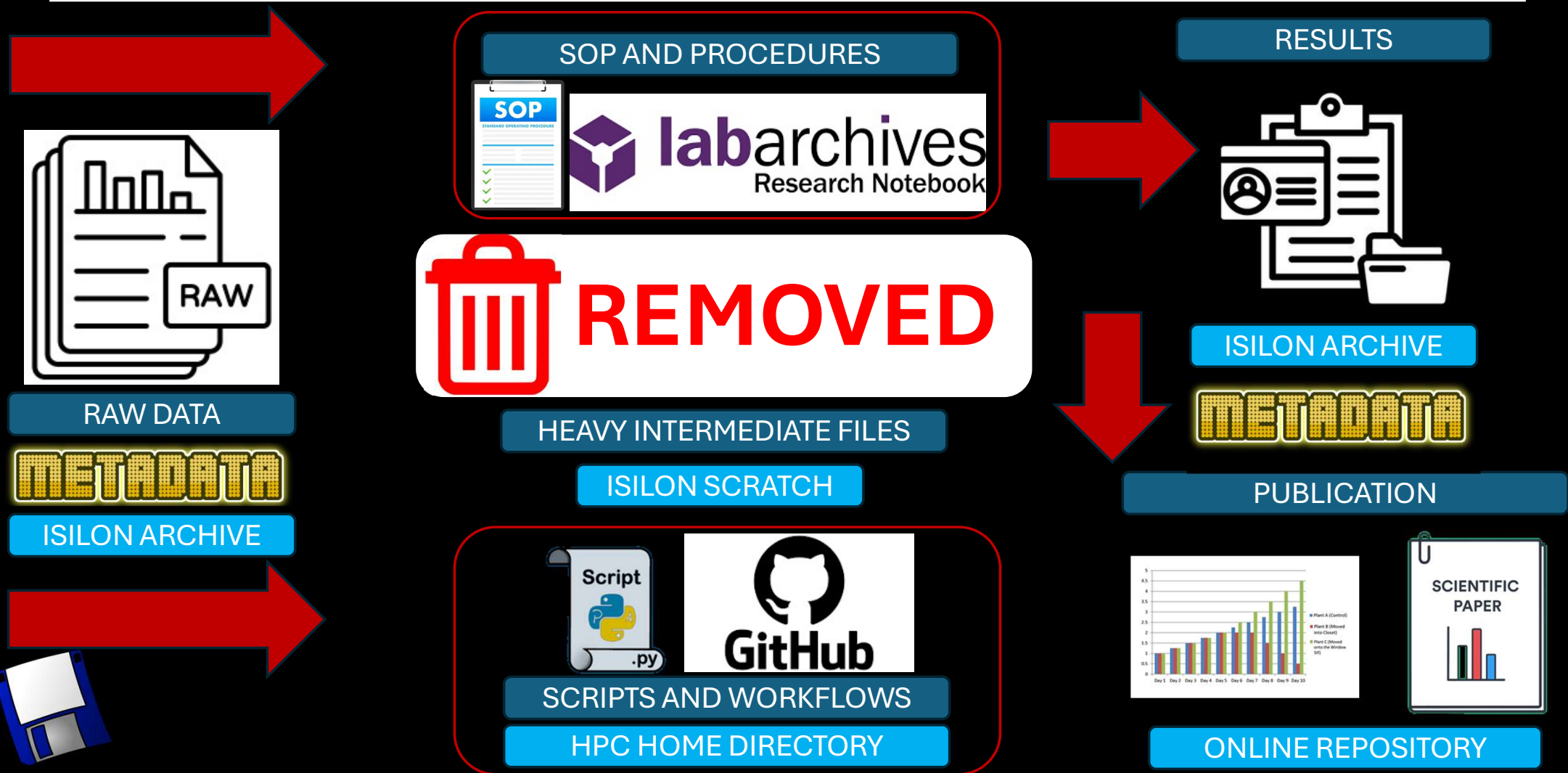


[New Data Centre]

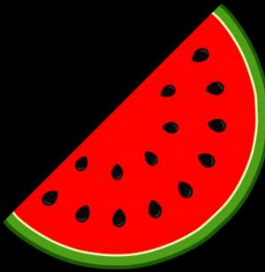
- NGI new data centre coming next year.
- This is a great opportunity to:
 - Reduce the footprint of our data storage.
 - Archive, organise, and properly annotate current datasets.
 - Review, standardise, and improve existing metadata.
 - Adopt and apply best practices for data generation, management, and storage.
 - Improve long-term data accessibility, reproducibility, and sustainability.
 - Prepare workflows and datasets for future scalability and collaboration.
 - And apply best practices going forward.



[Ideal Data Pipeline]



[Questions?]



Data Management

Part 2

Hands-on training



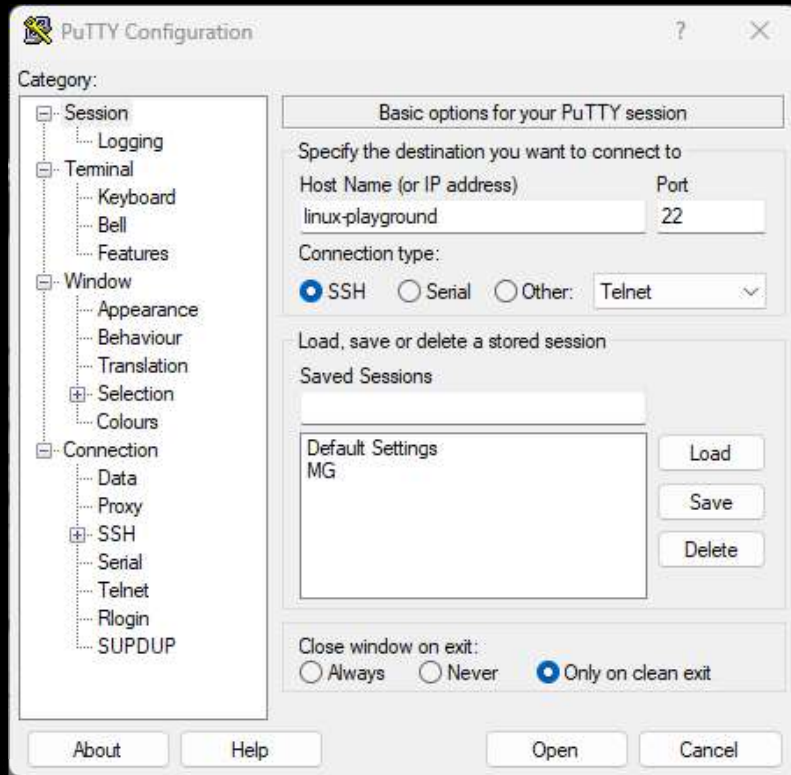
[The command line]\$

- Computers only speak in 0 and 1, but we use layers of abstraction to interact with them.
- From assembly to GUI, different layers simplify programming and system control.
- Command line tools provide direct access to system resources, making tasks faster and more efficient.
- Essential for automation, scripting, and HPC, where GUI access is limited.
- Commands can be combined (piped) to create powerful workflows, boosting productivity.



[Linux-playground]

- Using PuTTY to access HPC or VMs (linux-playground)



Please be aware, this is an end-user managed system.

Miguel Gonzalez Sanchez of JIC has management responsibility over the service on this machine.

Please contact the above in the first instance for support;

Alternatively contact the institute representative for Scientific Computing or the Research Computing (RC) team.

This machine has core services centrally controlled, via Puppet;
Any amendments made to controlled settings may revert automatically.

Date: Thu 13 Feb 17:04:28 GMT 2025
Hostname: v1373
CPU: 4 Cores, Load Average: 0.88, 0.83, 0.88
Memory: 3.7G Total, 681M Free

THIS SYSTEM IS NOT FULLY CONNECTED TO INTERNET.
Please contact Research Computing for further assistance.

```
-----  
o  o  o  o  o  o  o  o  o  o  o  o  o  o  o  o  o  o  o  o  o  o  
|  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  
|  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  
o  o  o  o  o  o  o  o  o  o  o  o  o  o  o  o  o  o  o  o  o  o  
-----
```

----- RESEARCH COMPUTING NBI 2025 -----

```
[luk24miz@v1373 ~]$ █
```

[Setting up today's lab]\$

- Open the browser in your workstation and create a folder in your U Drive named "informatics".
- After ssh into linux-playground run these two commands:
 - mountu will mount the recently created informatics folder in the /home/username/u_drive directory.
 - atraining will attach your screen to the presenter's ssh session.
- Open another ssh session using PuTTY or termina and ssh into linux-playground.
- The folder informatics from your U drive should be linked with the u_drive in your home directory, create a text file in Windows and check if is visible in your u_drive directory.




```

1 pwd
2 ls
3 cat file.tx
4 cd ~
5 ls -la
6 ls -lah

```

```

drwxr-xr-x   3 luk24miz JIC_a5   19 Nov 21   2024 .java
-rw-----   1 luk24miz JIC_a5   20 Apr 23 13:50 .lessht
t
-rwxr-xr-x   1 luk24miz JIC_a5  212 May 30   2024 ._SALMO
N-v.2.2.1
drwx-----   2 luk24miz JIC_a5   71 Sep 12 12:38 .ssh
drwxr-xr-x   5 luk24miz JIC_a5   75 Jul 21 12:23 .tg
v
-rw-r--r--   1 luk24miz JIC_a5  7.9M Jul 20 00:17 tgv-x86
_64-linux-musl.tar.gz
drwxr-xr-x   2 luk24miz root      0 Jul 21 11:57 u_drive
drwxr-xr-x   2 luk24miz JIC_a5   24 Sep 15 17:03 .vim
-rw-----   1 luk24miz JIC_a5  11K Sep 15 17:19 .viminf
o
-rw-r--r--   1 luk24miz JIC_a5  440 Jul 21 11:31 .wget-h
sts
-rw-r--r--   1 luk24miz JIC_a5  658 Oct 30   2024 .zshrc
7 $ cat

```

```

luk24miz@v1373:~
login as: luk24miz
Server refused our key
luk24miz@linux-playground's password:
Last login: Wed Feb 26 09:49:06 2025 from 149.155.218.43

-----
0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0
| 0 0 | 0 0 | 0 0 | 0 0 | 0 0 | 0 0 | 0 0 | 0 0 | 0 0 | 0 0 |
| 0 0 | 0 0 | 0 0 | 0 0 | 0 0 | 0 0 | 0 0 | 0 0 | 0 0 | 0 0 |
0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0
-----

##### Please be aware, this is an end-user managed system. #####

-----
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Alternatively contact the institute representative for Scientific
Computing or the Research Computing (RC) team.

This machine has core services centrally controlled, via Puppet;
Any amendments made to controlled settings may revert automatically.

Date: Wed 26 Feb 09:49:33 GMT 2025
Hostname: v1373
CPU: 10 Cores, Load Average: 1.11, 0.80, 0.40
Memory: 19G Total, 17G Free

THIS SYSTEM IS NOT FULLY CONNECTED TO INTERNET.
Please contact Research Computing for further assistance.

-----
0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0
| 0 0 | 0 0 | 0 0 | 0 0 | 0 0 | 0 0 | 0 0 | 0 0 | 0 0 | 0 0 |
| 0 0 | 0 0 | 0 0 | 0 0 | 0 0 | 0 0 | 0 0 | 0 0 | 0 0 | 0 0 |
0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0
-----

----- RESEARCH COMPUTING NBI 2025 -----

[luk24miz@v1373 ~]$

```

[Prompt, Home and \ (root)]

- Prompt (\$ or #) – The start of the command-line interface.
- Home Directory (/home/username/) – Each user's personal space.
Shortcut: ~ (tilde).
- Root Directory (/) – The top-level directory, containing all system files.
- root user owns / (everything) and has its home in /root
- Key System Folders – /bin (commands), /etc (configs), /home (users), /root (admin), /var (logs).



[Command structure]

```
$command -o --options argument1 argument2 argumenx ...
```

command application that we are calling.

-o --option these are the flag or options that we are using (can be grouped).

arguments these are the files or directories the command will use.

```
$cp -r dir1/file1.txt dir2/file1.txt
```



[Moving around]

- pwd print working directory
- ls list content of current directory (flags -l -h -r)
- cd change into specified directory
- cd .. takes you to the previous level
- cd ~ takes you to your home directory
- cd - takes you to the previous directory you were at



[Exercise 1]

- Change directory to the root of the filesystem.
- You can use `ls` to list files in the root directory.
- What happens if we add option `-l`?
- What if we use both `-l` and `-h`
- If you run the command with `-t` it will sort `ls` output by time of last change instead of alphabetically. `-r` will reverse the sort order. What happens if we use both at the same time?



[Relative and absolute paths]

- Relative paths:

../tsl/floor_1/proteomics_lab

./nbi/ei/floor_2/genomics_lab

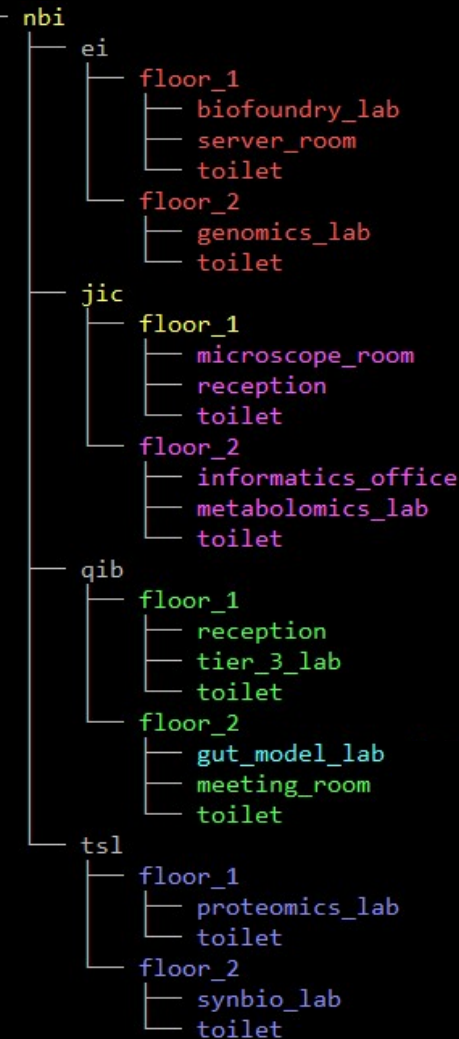
toilet

- Absolute paths:

/Norwich/nbi/jic/floor_1

/Norwich/

/Norwich/nbi/qib/floor2/toilet



[Exercise 2]

Type nbipath on the terminal and play the path game.

```
Current: /norwich/nbi/jic/floor_1
Quest 1/10 → /norwich/nbi/qib/floor_2/gut_model_lab
Required: relative path
Attempts left: 10
```

Available commands:

```
cd <path>      Change directory (relative or absolute)
pwd            Print current path
ls            List rooms and floors
tree          Show campus map tree
exit          Quit game
help          Show this help
```

Path types:

Absolute paths:

- Start with '/'
- Always point to the same location, no matter where you currently are
- Example:
 /norwich/nbi/jic/floor_1/toilet

Relative paths:

- Do not start with '/'
- Based on your current location
- Use:
 - . for current directory
 - .. for parent directory
- Example (from /norwich/nbi/jic/floor_2/office_A):
 ../floor_1/toilet



```
=====
NBI Navigation Game - by JIC Informatics
=====
Starting point: /norwich/nbi/jic/floor_1

===== NBI MAP =====
/norwich
├── nbi
│   ├── ei
│   │   ├── floor_1
│   │   │   ├── biofoundry_lab
│   │   │   ├── server_room
│   │   │   └── toilet
│   │   └── floor_2
│   │       ├── genomics_lab
│   │       └── toilet
│   ├── jic
│   │   ├── floor_1 # Miguel <<<----- YOU ARE HERE
│   │   │   ├── microscope_room
│   │   │   ├── reception
│   │   │   └── toilet
│   │   └── floor_2
│   │       ├── informatics_office
│   │       ├── metabolomics_lab
│   │       └── toilet
│   ├── qib
│   │   ├── floor_1
│   │   │   ├── reception
│   │   │   ├── tier_3_lab
│   │   │   └── toilet
│   │   └── floor_2
│   │       ├── gut_model_lab @ <<<----- YOUR DESTINATION
│   │       ├── meeting_room
│   │       └── toilet
│   └── tsl
│       ├── floor_1
│       │   ├── proteomics_lab
│       │   └── toilet
│       └── floor_2
│           ├── synbio_lab
│           └── toilet
└── ...
```

📍 Current: /norwich/nbi/jic/floor_1
🎯 Quest 1/10 → /norwich/nbi/qib/floor_2/gut_model_lab
🔑 Required: relative path
★ Attempts left: 10
💡 Type help if you are lost

[Creating, moving and copying]

- `mkdir` make a new directory
- `rmdir` remove an empty directory
- `cp` copy a source to a destination (`-r` for directories)
- `mv` move a source to a destination
- `rm` remove a file (`-r` for directories)
- `touch` update the access date and/or modification date of a file,
but if it does not exist the file is created, quite useful
sometimes to reserve a filename.



[Exercise 3]

- Inside your u_drive directory create a new directory named house.
- Inside the house directory create a few directories named living-room, toilet, dining-room and kitchen
- Create a file named chair.file inside the toilet, then move it to the dining-room.
- Copy the chair.file from the dining-room to the kitchen.
- Create a new file inside the kitchen named shower.file, after that copy the shower.file into the toilet and then remove the shower.file from the kitchen.
- Create more rooms/furniture and move and copy them around the house.



[rsync]

Unlike `cp`, which copies files in full each time, `rsync` detects and transfers only the parts of files that have changed. This makes it significantly faster and more efficient for repeated or large-scale syncs. Additionally, `rsync` supports powerful features and resume capabilities, which `cp` does not.

- Works locally or over SSH for remote sync.
- Supports exclusions, compression, and dry-run options.
- Commonly used for backups and mirroring
- Basic flags:
 - a (archive): Enables recursive copy and preserves symbolic links, permissions, timestamps, group, and owner.
 - v (verbose): Shows detailed output of what `rsync` is doing.
 - z (compress): Compresses data during transfer (useful for slow network connections).



[rsync]



[rsync]



[rsync – Command Structure & Examples]

- Basic Syntax:
 - `rsync [options] source destination`
- Examples:
 - Local sync:
 - `rsync -av /data/source/ /backup/data/`
 - Remote sync over SSH:
 - `rsync -avz /data/ user@remote:/backup/`
 - With exclusions:
 - `rsync -av --exclude '*.tmp' /data/ /backup/`
 - Dry run (preview changes):
 - `rsync -av --dry-run /data/ /backup/`



[Working with compressed data]

Bioinformatics files are huge, and compression reduces storage needs, speeds up transfers, and minimises I/O bottlenecks. Many tools can process compressed files directly, improving efficiency.

- gzip - compress a file and adds .gz extension.
- gunzip - decompress .gz files.
- zcat - view or concatenate compressed files without decompressing.
- tar - archive multiple files into one (a tarball).
- Tar examples:
 - tar -cvf archive.tar file1 file2 → create a tarball containing file1 and file2
 - tar -xvf archive.tar → extract contents from archive.tar
 - tar -czvf archive.tar.gz dir/ → compress a directory into a .tar.gz archive
 - tar -xzvf archive.tar.gz → extract a .tar.gz archive



[Exercise 4]

- Exercise 1: copy the folder `compressing_exercise` using `rsync` from `/` into your `u_drive` directory in your home.
- Exercise 2: Compress `reference.fastq` using `gzip`.
- Exercise 3: View the contents without decompressing.
- Exercise 4: Uncompress the file.
- Exercise 5: compress the `data/` folder. Check the size using `du -sh data/`
`data_archive.tar.gz`.
- Exercise 6: Remove the current `data/` folder and extract it from the archive.



[md5sum – File Integrity Check]

- Calculates a 128-bit hash (checksum) for files.
- Used to verify file integrity after transfers or backups.
- Simple to use:

`md5sum filename` (can be piped to a file > checksum.txt).

- Compare with expected checksum:

`md5sum -c checksums.txt`

- Note: Recompressing or uncompressing a file changes its MD5 – even if the original content remains the same.



[Exercise 5]

- Create the md5sum from reference.fastq.
- Copy the file and get the md5sum from the new file and compare it to the old one.
- Add a space to the file and get the md5sum, compare it to the original.
- Remove the space and get again the md5sum. What has happened?



[Find]

- Search for files and directories based on name, type, size, or other attributes.
- `find [path] [options] [expression]`
- Find by name: `find /home/user -name "file.txt"`
- Find by type: `find /var -type d` (directories) / `find /var -type f` (files)
- Find by size: `find / -size +100M` (files larger than 100 MB)

- Tips:

- `.` → current directory

- `-iname` → case-insensitive name search

- `-mtime` → files modified in the last N days

- Examples:

- `find . -name "*.fastq.gz"`

- `find . -name "*.fastq"`

- `find . -type f -size +100M`

- `find . -type f -size +1M`

- `find . -type f -mtime +5000`



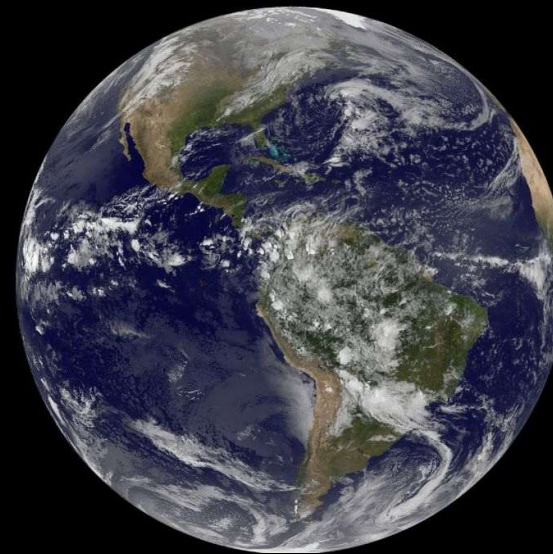
[Exercise 6]

- Copy /datamanagement to your ~/u_drive folder.
- Find uncompressed fasta files in there and compress them.
- Find any file bigger than 100MB
- Find the files older than a year in the original /datamanagement directory

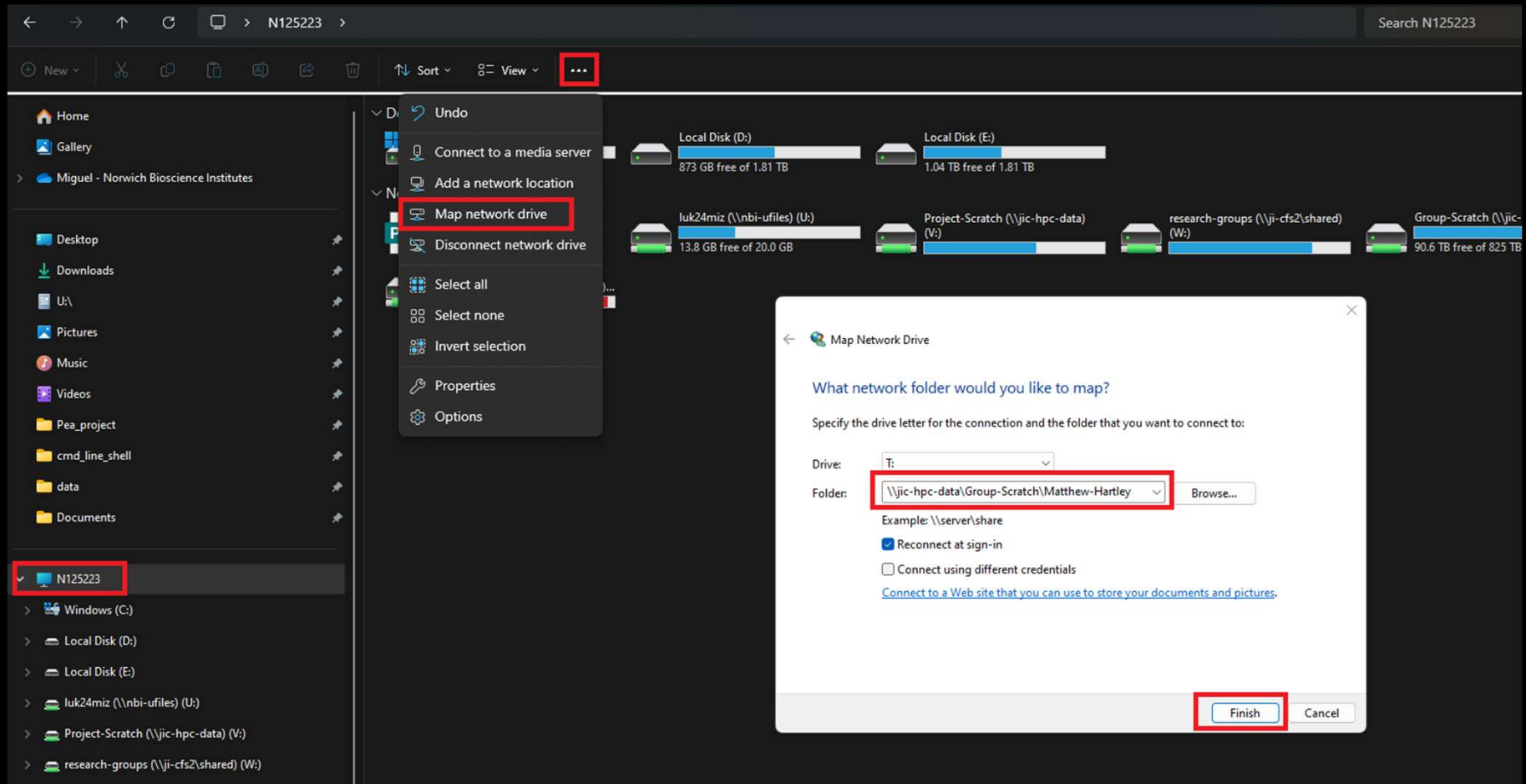


[Big data → HPC Storage]

- So far, the scope has been restricted to small datasets, but the aim of this course is to handle large sequences, big batches of images... in other words, big scientific data.
- As mentioned earlier, the data should be stored on the NBI HPC storage.
- Research Computing manages between all of the institutes' resources more than 40PB of data storage across over 3500 disk drives, with an additional over 600TB used solely for performance.

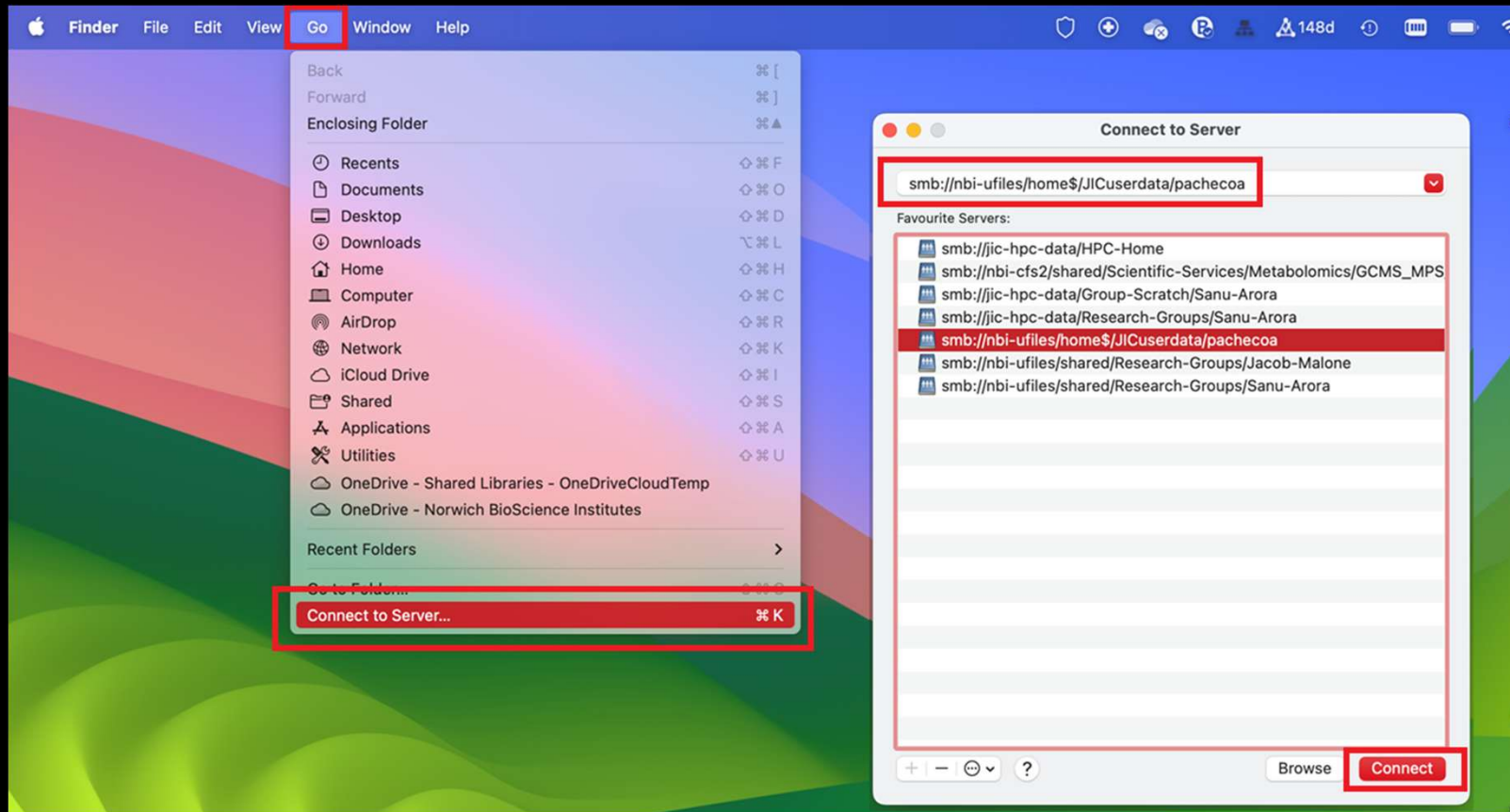


[Accessing HPC storage from Windows]



\\jic-hpc-data\HPC-Home for the home directory

[Accessing HPC storage from macOS]



[Accessing HPC storage from Linux]

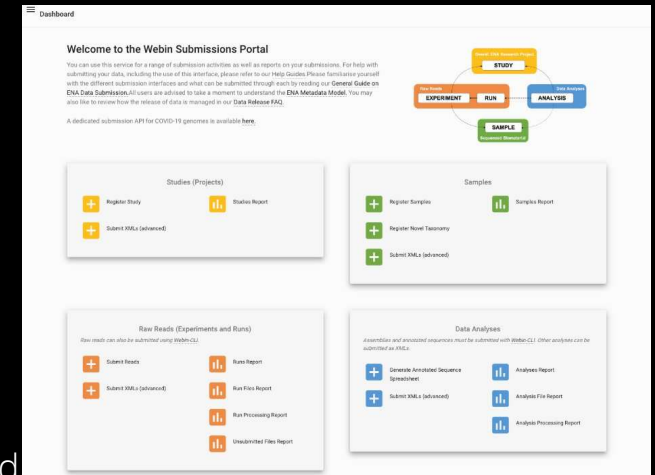
- As a Linux user you are brave enough and you are on your own.
- Jokes aside, you need sudo rights at least to run the “mount” command.
- This is an example on how to mount the project scratch folder:

```
luk24miz@n77864:~  
(base) [luk24miz@n77864 ~]$ sudo mount.cifs //jic-hpc-data/project-scratch projects/ -o domain=nr4,user=$(id -un),uid=$(id -u)  
[sudo] password for luk24miz: 
```



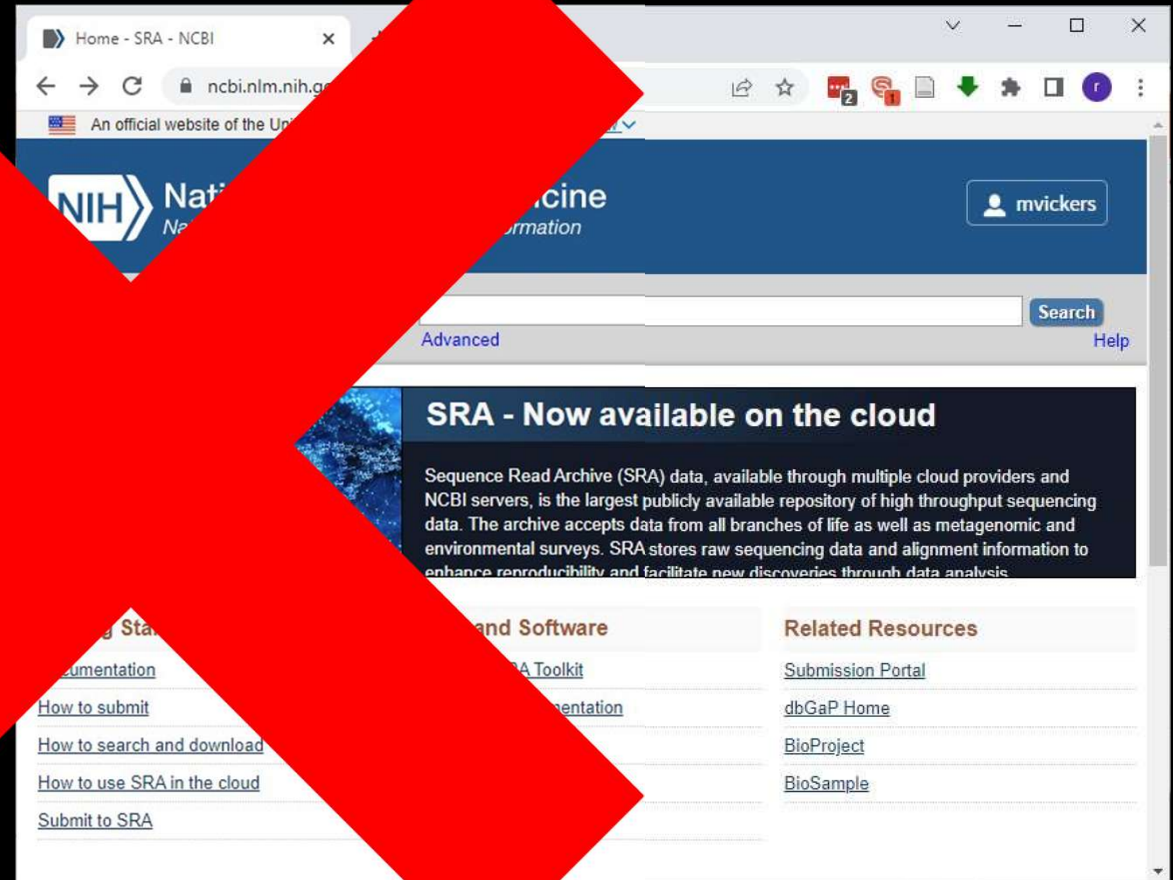
[Uploading to EBI]

- Two main systems of interest
- ENA – for sequence data
 - <https://www.ebi.ac.uk/ena/submit/webin/>
- Array Express
 - <https://www.ebi.ac.uk/biostudies/arrayexpress>.
 - For raw sequence data + processed results.
 - Array Express links processed data to the raw sequenced data
- <https://www.youtube.com/watch?v=ANr2PTVy7JA>
- Make sure your raw data file names contain only letters, numbers, dots and underscores. Nothing else.
- You will need to fill in a lot of metadata and upload using FTP.
- Support response time frame is 3 weeks currently – so start early!!



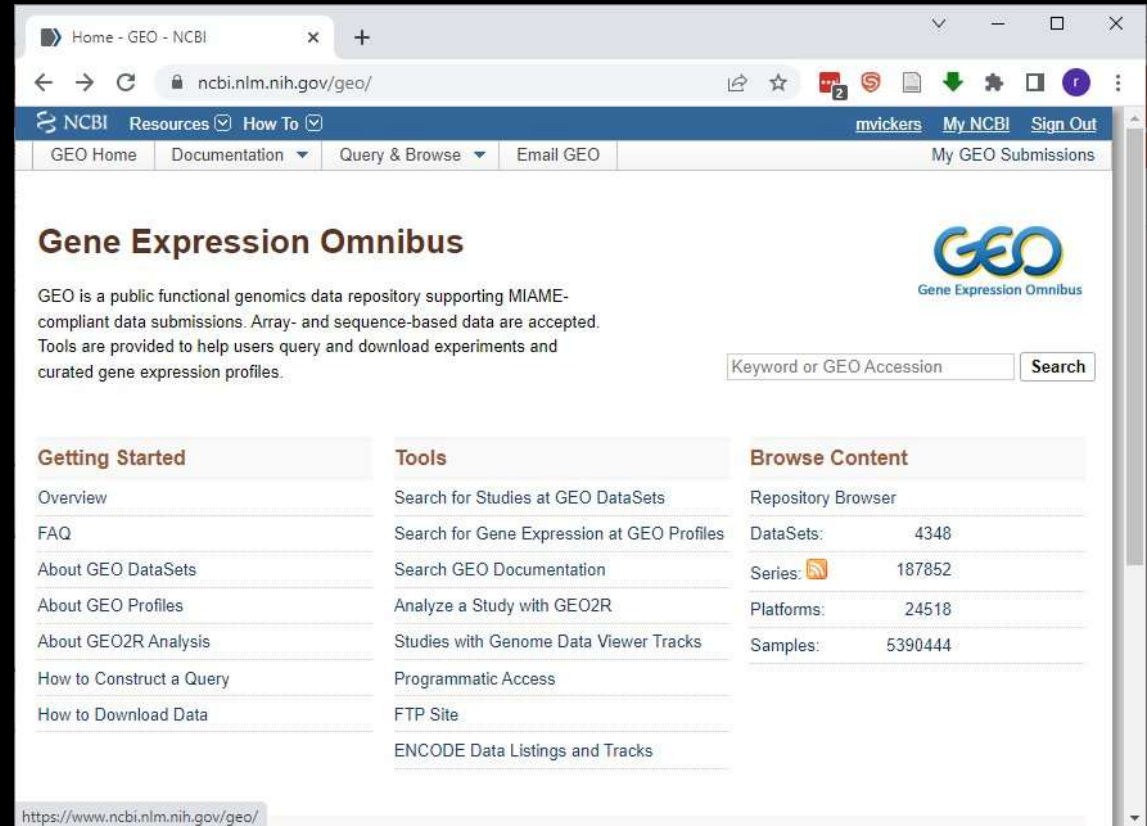
[Uploading to SRA]

- For RAW data.
- Perfect for NGS sequencing
- Enforces some basic metadata

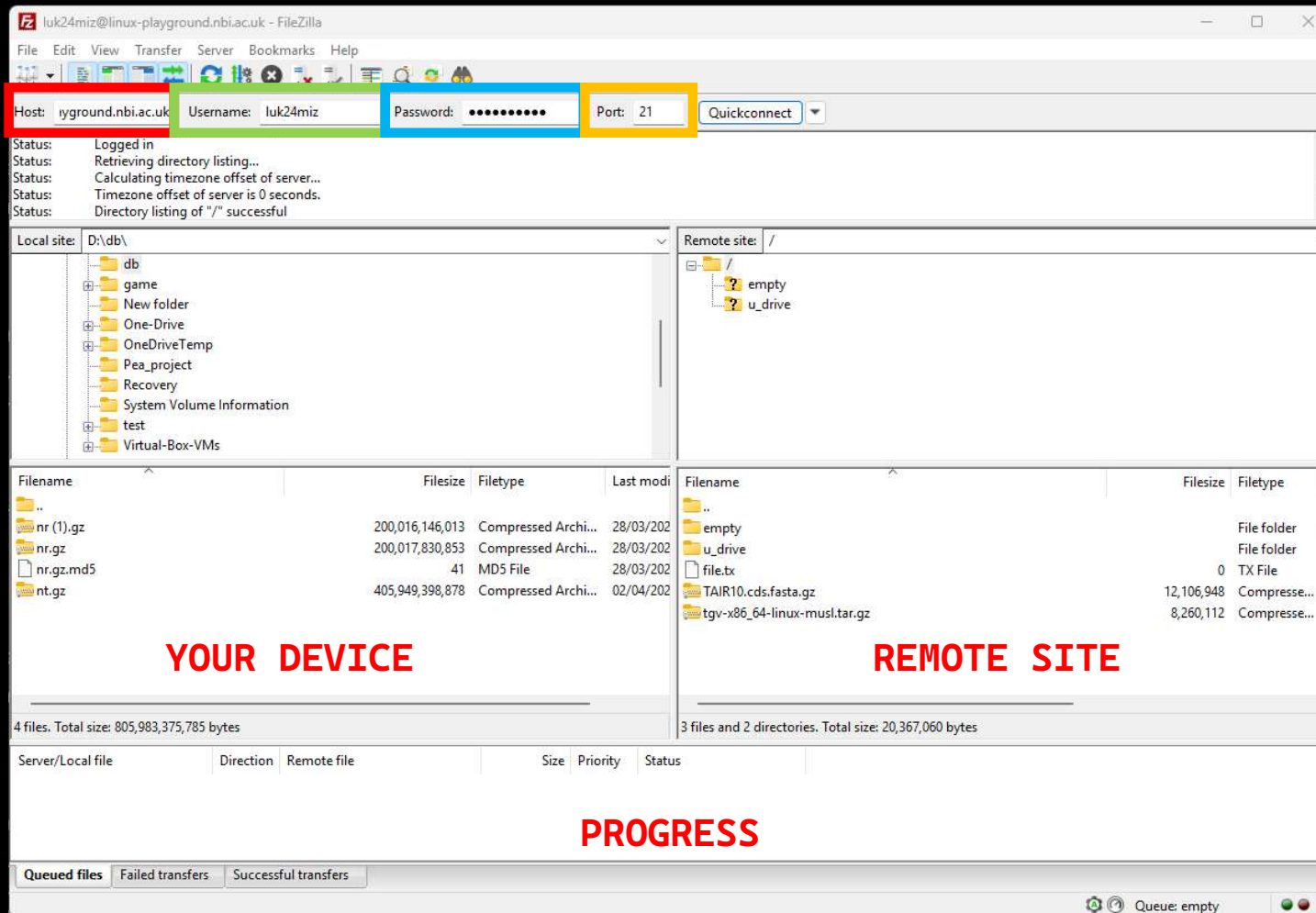


[Uploading to GEO]

- Great for NGS experiments.
 - RAW + processed data.
- Automatically uploads raw data to SRA.
- Pretty good at enforcing metadata.
- Queries often answered within a working day.
- Brief look at metadata sheet (included in your resources).



[FTP Filezilla]



Host

Username

Password

Port

[Downloading data into the HPC]

- Accessing the HPC Storage:

- Users must use their own devices as a gateway to interact with the HPC storage.
- External access from the HPC is limited, with only a few exceptions (online databases/repositories) permitted.

- Download partition:

JIC informatics, Research Groups, Data champions and designated users can download data into the HPC storage through the download partition.

- Sharing data in/out:

- Dropoff (<https://dropoff.nbi.ac.uk/>) up to 300GB, easy to use.
- Globus (<https://www.globus.org/>) No data limit.



[Exercise 7]

- Map your HPC-home folder.
- Create a file with your username in your U drive
- Use Filezilla to upload the file to the ftp version of linux-playground



[JIC archive – Data Champion]

- `jic-data-mover{01..06}` is accessible only by data champions and is the only way to write to `/jic/archive`.
- `/jic/{scratch,research-groups}` etc. read only on `jic-data-mover{01..06}` to prevent accidental deletion.
- `/jic/archive` is read only to group members across HPC.
- `/jic/archive` has snapshots.



[TSL Users]

- TSL has specific policies and metadata requirements:
- To find out more, you can visit:

https://bioinformatics.tsl.ac.uk/posts/workflows-and-literate_compute/

<https://bioinformatics.tsl.ac.uk/>

<https://intranet.nbi.ac.uk/cms/6741>

- But as I said earlier, please contact the TSL Bioinformatics teams if you have any doubts.



[Concluding Recommendations]

- Data integrity Quality Checks before analyses.
- Check that you've downloaded the data correctly.
- Keep raw data and derived data separate.
- Compress your sequence data.
- Keep redundancy low. Remove intermediate files. Regular cleanup.
- Use your resources wisely.
- Archive important data.
- Follow BioFAiR recommendations.
- Get your data ready for A.I.
- Ask, we are here to help.



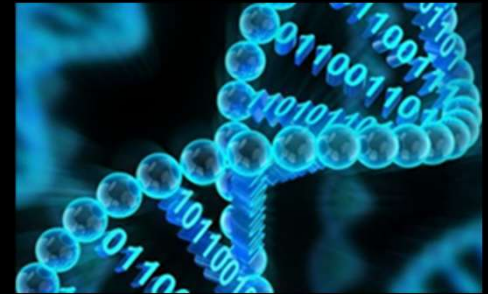
[Take home message]

- Know your institution's data policy.
- Talk to your informatics team early and often.
- Consider becoming a data champion in your group.
- Develop and align on common data practices within your group.
- Store data on resilient, backed-up infrastructure.
- Organise your metadata clearly and consistently.
- Keep your data safe. Annotate it. Don't waste storage resources.



[Bioinformatics Drop-in Sessions]

- Wednesdays at 11:00-12:00 in the Rec.
- Free Biscuits available.
- Come and ask questions/talk Bioinformatics:
- Some things we may be able to help with right away, some we may need to think about. We can do some thinking before the session.
- (and may give better answers) if you email us beforehand at informatics@jic.ac.uk



ALL WELCOME



[REC]

I organise the REC-ROTA. We are always welcoming volunteers to join us to help on Fridays at the bar, and we pay in drinks!

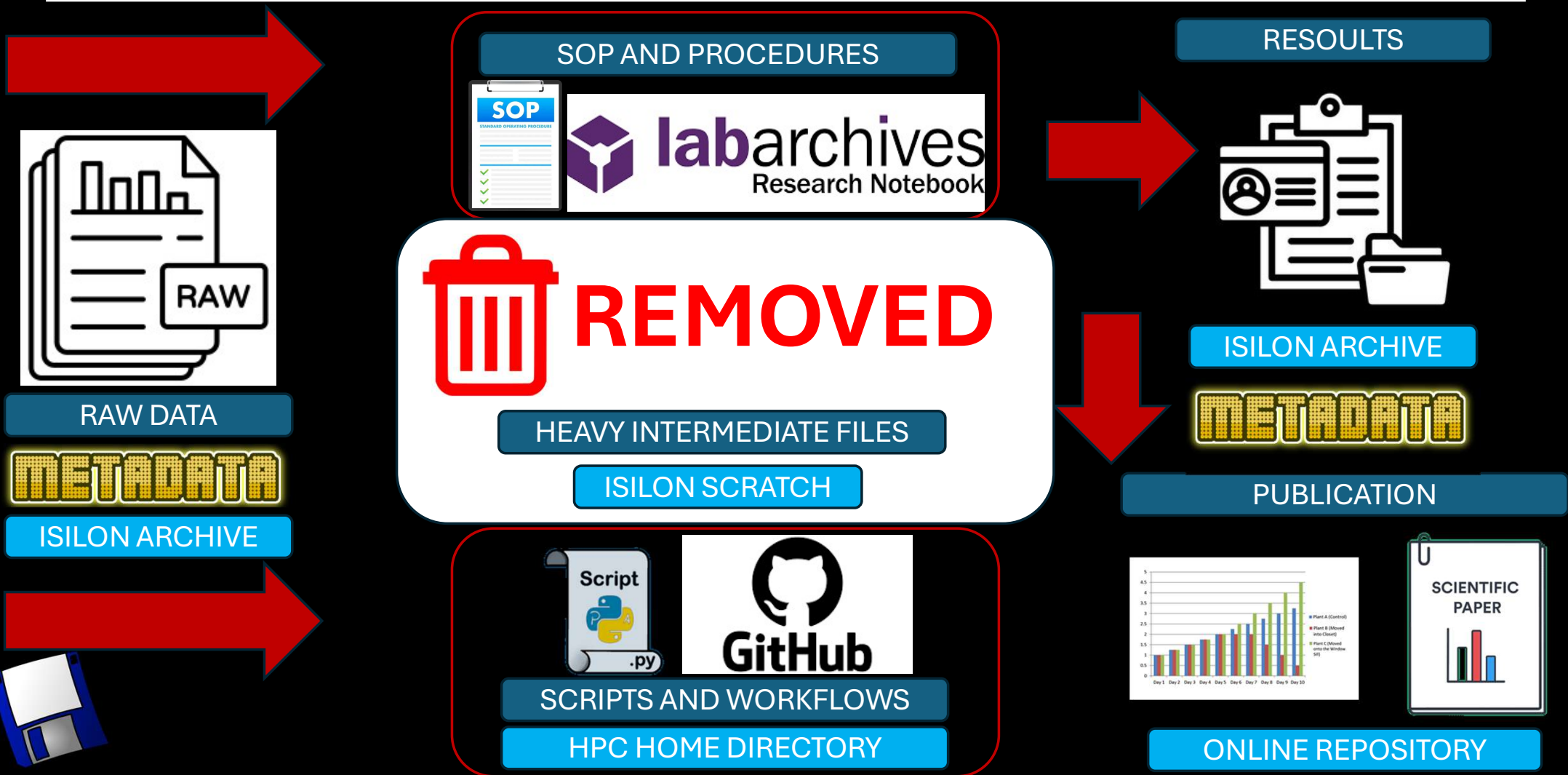


[Upcoming courses]

- 16-06-2026 Introduction to Python – Part 1.
- 17-06-2026 Introduction to Python – Part 2.
- 18-06-2026 Introduction to Python – Part 3.
- 23-09-2026 Invitation to AI.
- 14-10-2026 Bioinformatics and how to learn it.



[Ideal Data Pipeline]



[Questions?]

